



Koneru Lakshmaiah Education Foundation

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OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026-27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.:12

Team No.:26

Project Title: KERNEL MONITOR

Team Members

Roll Number	Student Name	Signature
2520090184	R Sai Charan Reddi	
2520090100	G Mani Deep Patel	
2520090155	P Akhil Yadav	

1. Abstract

Kernel Monitor is a Linux-based system monitoring project designed to observe and display important operating system and kernel-level information in a user-friendly manner. The project focuses on understanding how the Linux kernel manages system resources such as CPU, memory, processes, and system information. It uses Linux system interfaces, system calls, and the `/proc` virtual filesystem to collect and analyze real-time system information. The proposed system monitors CPU utilization, memory usage, running processes, process IDs, system uptime, and other relevant kernel information. The collected data is processed by a C/C++ based monitoring application and presented in a clear and understandable format. Users can refresh the monitor to observe changes in system resource utilization and process activity. The project provides practical understanding of important Operating Systems concepts including process management, CPU scheduling, memory management, system calls, process identification, and interaction with the Linux kernel. It also demonstrates how user-level programs can obtain system information through Linux-provided interfaces. The expected outcome is a lightweight Kernel Monitor that helps users observe system resources and understand the interaction between user programs and the Linux operating system kernel.

2. Problem Statement

- 1.Modern operating systems continuously manage CPU resources, memory, processes, and other system activities in the background. However, understanding how these resources are utilized at the operating-system level can be difficult for students without practical observation tools.
- 2.The **Kernel Monitor** project aims to develop a Linux-based monitoring application that collects and displays important system and kernel-related information such as CPU utilization, memory usage, running processes, process IDs, and system uptime.

3. The project is relevant to Operating Systems and Systems Programming because it provides practical implementation of process management, memory management, system information retrieval, Linux system interfaces, and system calls. It helps users understand how applications interact with the Linux kernel to obtain information about the current state of the system.

3. Objectives

1. To develop a Linux-based application for monitoring important system and kernel-level information.
2. To monitor CPU utilization, memory usage, system uptime, and running processes in real time.
3. To understand and implement Linux system calls, POSIX APIs, and the `/proc` filesystem for retrieving system information.
4. To provide practical understanding of Operating Systems concepts such as process management, memory management, CPU utilization, and interaction between user-level programs and the Linux kernel.

4. Proposed Methodology

1. The Kernel Monitor will be designed and implemented as a Linux-based C/C++ application. The application will obtain system information from Linux kernel interfaces and system calls.
 2. First, the program will identify the current system and process information. It will access relevant information provided by the Linux `/proc` virtual filesystem, such as CPU, memory, uptime, and process information. Linux system calls and POSIX APIs will also be used wherever required to obtain process-related information.
 3. The collected information will be processed by the application to calculate or display CPU utilization, memory utilization, process details, process IDs, system uptime, and other system information.
 4. The monitor will provide a simple interface through which users can view the current system status and refresh the information periodically. The overall flow is:
5. **Linux Kernel → `/proc` / System Calls → Kernel Monitor Application → Data Processing → System Information Display**

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API / System Call	Purpose in the Project
Process Management	To monitor running processes and their states
Process Identification	To obtain and display Process IDs (PIDs)
<code>/proc</code> Filesystem	To obtain CPU, memory, uptime, and process information
<code>fork()</code>	To understand/process creation where required
<code>getpid()</code>	To obtain the PID of the current monitoring process
<code>getppid()</code>	To obtain the parent process ID

sysinfo()	To obtain system-level information such as memory and uptime
open()	To open required system/proc files
read()	To read information from system/proc files
close()	To close opened files and release resources
opendir()	To access process directories under /proc
readdir()	To read process directory entries
closedir()	To close opened directories
POSIX APIs	To interact with Linux operating-system services
CPU Utilization	To observe CPU resource usage
Memory Management	To monitor available and used memory

6. Individual Contribution

Roll Number	Student Name	Individual Responsibility
2520090184	R Sai Charan Reddi	Kernel Monitor design and project coordination Testing, documentation and GitHub management
2520090100	G ManiDeep Patel	CPU and memory monitoring module
2520090155	P Akhil Yadav	Process monitoring and /proc filesystem implementation

7. Tools / Platforms / Software Used

Tool / Platform / Software	Purpose
Linux / Ubuntu	Operating environment for implementing and testing the Kernel Monitor
C / C++	Main programming language for system-level implementation
GCC / G++	Compilation of C/C++ source code
Linux Terminal	Compiling, executing and testing the application
/proc Filesystem	Retrieving kernel and system information
POSIX APIs	Accessing operating-system services
Git / GitHub	Version control and project repository
VS Code / Vim / Nano	Source-code development and editing

8. Expected Outcome

The expected outcome of the project is a functional Linux-based **Kernel Monitor** capable of displaying important system and kernel information.

The application will provide information such as CPU utilization, memory usage, system uptime, running processes, process IDs, and other relevant system details. The monitor will retrieve information using Linux kernel interfaces, the /proc filesystem, system calls, and POSIX APIs.

The project will demonstrate practical knowledge of Operating Systems concepts including process management, memory management, system information retrieval, system calls, and user-level interaction with the Linux kernel.

The final application will be lightweight, easy to execute from the Linux terminal, and useful for students to understand how operating-system resources are monitored and managed.

9. GitHub Repository

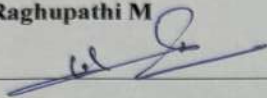
GitHub Repository Link: https://github.com/gundamanideep-dot/KLH_CSIT_2026_26_Kernal_monitor

Repository Created: Yes

☐ Yes ☐ No

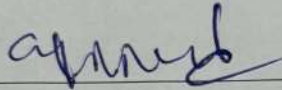
Faculty Name and Signature

Faculty Name: Raghupathi M

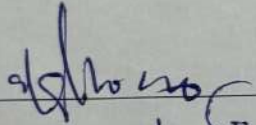
Signature: 

Remarks: _____

Faculty Use

Project Approved: 

☒ Yes ☐ No

Date: 

Faculty Signature: 